Brice Peres

Computer Graphics & Simulation Engineer

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Portfolio: **bricEprs.github.io** ↗

Experience

Inria Elan team – PhD candidate **Apr. 2026 – present**
Numerical study of naturally-curved fibre assemblies *Grenoble, France*

- Numerical modelling and simulation of bundles of naturally-curved fibres (continuation of the work below).

Inria Elan team – Pre-PhD research intern **Sept. 2025 – Mar. 2026**
Tomographic hair-data filtering *Grenoble, France*

- Designed filtering methods for tomographic acquisitions of human hair, in preparation of the PhD project.

Dassault Systèmes – End-of-studies intern **Sept. 2024 – Mar. 2025**
3D engine performance optimisation *Paris, France*

- Worked with the rendering team to improve the performance of one of Dassault Systèmes' main 3D engines.

Corys – R&D intern **May 2023 – Sept. 2023**
Unreal Engine 5, runtime train simulator *Grenoble, France*

- Implemented real-time windscreen effects (rain, snow, dirt, cleaning) on Unreal Engine 5 (HLSL).
- Worked in the R&D team led by Alain Kocelniack.

Inria – Research intern **June – July 2022**
GPU terrain filtering *Grenoble, France*

- Designed a GPU height-map filtering algorithm for terrain generalisation (advisors: Joëlle Thollot, Romain Vergne).

Selected projects

Real-time soft-body simulation **Dec. 2024**
C++, CUDA
Spring-mass soft body accelerated on the GPU with CUDA.

Progressive raytracer **Dec. 2024**
C++, OpenGL
GPU raytracer with BVH traversal, shadows, reflections, refractions and environment maps. Cook–Torrance shading with a GGX microfacet distribution; frame accumulation for progressive refinement.

Particle physics simulation **May 2023**
C++
2D/3D simulation under gravity and a Lennard-Jones potential with custom spatial data structures. Ran 8 000 particles over 400 000 steps in 20 minutes on a laptop.

Black Seas – multiplayer naval game **Feb. – Apr. 2023**
Rust, OpenGL – team of 5
From-scratch engine; rendering, terrain, lighting, post-processing (ray-marched clouds, bloom, tonemapping).

Education

ENSIMAG – Engineering degree (M.Eng.), Grenoble INP **2021 – 2024**
Computer science and applied mathematics *Grenoble, France*

Institute of Science Tokyo (formerly Tokyo Tech) **Mar. – Aug. 2024**
Academic exchange semester *Tokyo, Japan*

Lycée Notre-Dame de Sion – Classes préparatoires MPSI/MP **2018 – 2021**
Two-year intensive course for French engineering school exams *Marseille, France*

Skills, languages & interests

Languages. C++, Rust, C, Python, GLSL, HLSL
Graphics & GPU. OpenGL, CUDA, WebGL, Unreal Engine 5, Shadertoy
Tools. Git, LaTeX, Linux • **Familiar.** Java, Shell, Assembly
Spoken. French (native) • English (C1, TOEFL 100 pts) • Japanese (A1)
Interests. Climbing & bouldering since 2011 (a few competitions); computer graphics, physically based rendering, real-time simulation, Shadertoy.